THE MALAWI NATIONAL EXAMINATIONS BOARD

2006 MALAWI SCHOOL CERTIFICATE OF EDUCATION EXAMINATION

MATHEMATICS

Subject Number: M131/II

Time Allowed: 2 h 30 mins

Monday, 23 October **8:30 – 11:00 am**

PAPER II

(100 marks)

Instructions

1. **This paper contains 6 pages. Please check.**
2. Answer **all** the **six** questions in Section **A** and any **three** questions from Section **B**.
3. The maximum number of marks for each answer is indicated against each question.
4. Mathematical tables, graph paper and answer books are provided.
5. Calculators may be used.
6. Used graph paper and/or supplementary sheets must be tied together inside the answer book with a string.
7. **All working must be clearly shown**; it should be done on the same sheet as the rest of the answers.
8. Write your **Examination Number** on top of each page of your Answer Book.

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Section A (55 marks)

Answer **all** the **six** questions in this section.

1. a. Simplify $\left(\sqrt{2}+\sqrt{3}\right)\left(\sqrt{8}-\sqrt{12}\right)$ without using a calculator and four-figure tables. **(4 marks)**

b. Given that $f(x) = x + 3$ and $f(2) = 2 + x + g(x)$, find $g(x)$. **(4 marks)**

2. a. Evaluate $2\log_a a + 3\log_a 1 + \log_2 \sqrt{2}$. **(4 marks)**

b. **Figure 1** shows a straight line graph.

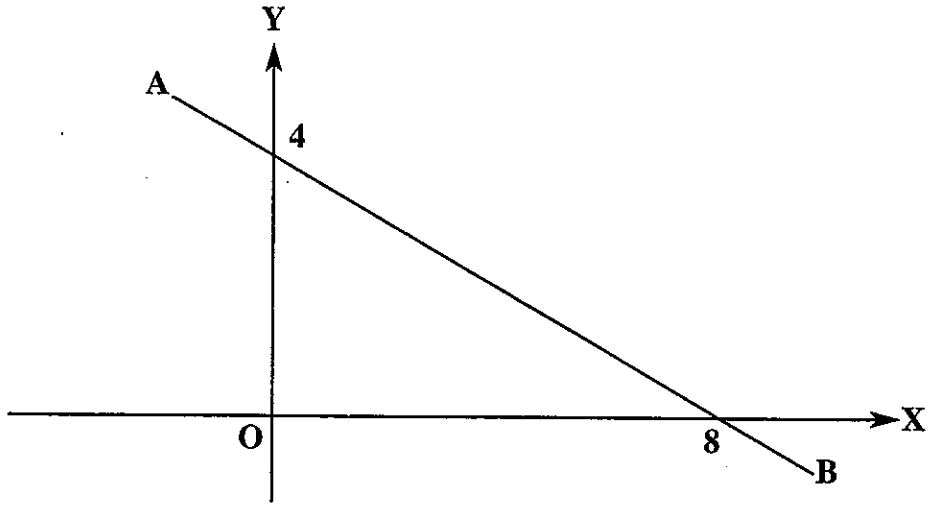

Figure 1

Find the equation of line **AB** and write it in the form $y = mx + c$. **(4 marks)**

3. a. A coin is tossed three times. Calculate the probability that the first and the third tosses give heads as outcomes, regardless of what the outcome of the second toss is. **(4 marks)**

b. A person running at a speed of y metres per second passes a stationery train 50 metres long. If the person takes $\frac{25}{y-1}$ seconds to pass the train, calculate the value of y. **(4 marks)**

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4. a. Given that $\binom{a-1}{2} + \underline{b} = \binom{16}{7}$. Find $\underline{b}$. **(3 marks)**

b. A motorist accelerates uniformly from rest to 40 m/s in 8 seconds. He then accelerates uniformly to 90 m/s in the next 6 seconds.

(i) Using a scale of 2 cm to represent 20 m/s on the vertical axis and 2 cm to represent 2 seconds on the horizontal axis, draw a speed-time graph for the first 14 seconds.

(ii) Calculate the acceleration for the last 6 seconds. **(7 marks)**

5. a. Blouses and shirts for school uniforms were selling at K500 and K250 respectively. Phiri family bought two blouses and four shirts, while Mwale family bought three blouses and six shirts for their children.

(i) Present this information in two matrices.

(ii) Using matrix multiplication, calculate the amount of money each family spent on the clothes. **(5 marks)**

b. Using a ruler and pair of compasses only, construct, in the same diagram,

(i) a circle centre **O** of radius 3 cm.

(ii) two tangents to the circle from a point **T** which is 8 cm from **O** touching the circle at points **M** and **N**.

(iii) Measure and state angle **MTN**. **(6 marks)**

6. **Figure 2** is a circle **ABDC** centre **O**. **TD** is a tangent, **TBA** is a straight line and **AD** is a diameter.

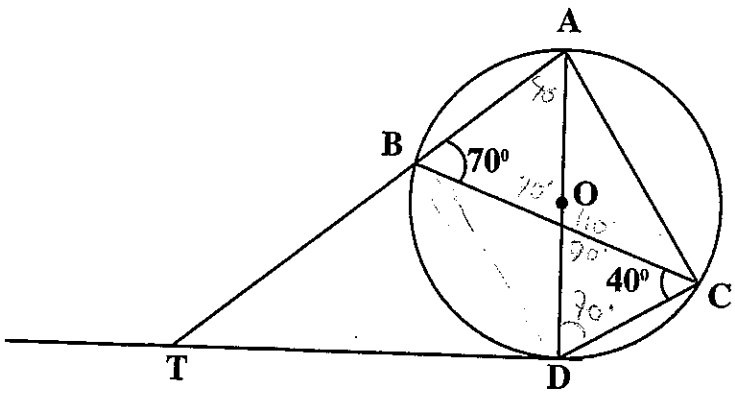

Figure 2

If angle **ABC** = 70° and angle **BCD** = 40°, calculate:

a. angle **BDC**; **(5 marks)**

b. angle **ATD**. **(5 marks)**

Continued/...

Section B (45 marks)

Answer any **three** questions from this section.

7. a. The areas of two similar triangles are 64 cm^2 and 36 cm^2. The base of the bigger triangle is 16 cm. Calculate the corresponding height of the smaller triangle. **(5 marks)**

b. Solve the equation $2x^3 - 5x^2 + x + 2 = 0$. **(10 marks)**

8. a. Make y the subject of the formula $y^{\frac{1}{2}} = \dfrac{w + w(xy)^{\frac{1}{2}}}{x^{\frac{1}{2}}}$. **(5 marks)**

b. An aeroplane leaves airport **A** on a bearing of N23°E and flies for 340 km to another airport **B**. It then leaves airport **B** and flies on a bearing N60°W to another airport **C**. If the airports **A** and **C** are 680 km apart, calculate the bearing of airport **A** from airport **C**. **(10 marks)**

9. a. A church congregation has a youth group and music group. There are 400 people in the congregation out of which 40 people belong to both the youth group and music group. There are 60 members who belong to the youth group only, while 220 belong to neither the youth group nor the music group.

(i) Draw a venn diagram illustrating this information.

(ii) Calculate the number of people who belong to the music group only. **(5 marks)**

b. A farmer gets a loan of K50 000 to buy sheep and goats only. Sheep cost K5 000 each while goats cost K2 000 each. She would like to spend at least K10 000 more on sheep than on goats.

(i) If x represents the number of sheep and y represents the number of goats, write down two inequalities in x and y in addition to $x \geq 0$ and $y \geq 0$.

(ii) Using a scale of 2 cm to represent 2 units on the horizontal axis and 2 cm to represent 5 units on the vertical axis, draw the region which represents the four inequalities.

(iii) How many sheep and goats can the farmer buy to have the maximum number of animals with the loan. **(10 marks)**

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1. a. **Table 1** shows tallies of students' marks in a test.

Table 1

MARK (m)	TALLY
$0 \leq M \leq 10$	
$10 < M \leq 20$	//
$20 < M \leq 30$	///
$30 < M \leq 40$	///
$40 < M \leq 50$	~~////~~ ~~////~~
$50 < M \leq 60$	//
$60 < M \leq 70$	~~////~~ ////
$70 < M \leq 80$	~~////~~ /
$80 < M \leq 90$	////
$90 < M \leq 100$	/

(i) Find the modal class.

(ii) If 50% of the students were to pass, find the passing mark for the test. **(7 marks)**

b. **Figure 3** shows a wooden prism with triangular cross-section **ABC**. **BE** = 20 cm, **AB** = **BC** = **AC** = 6 cm, **ACFD, BCFE** and **ADEB** are rectangles.

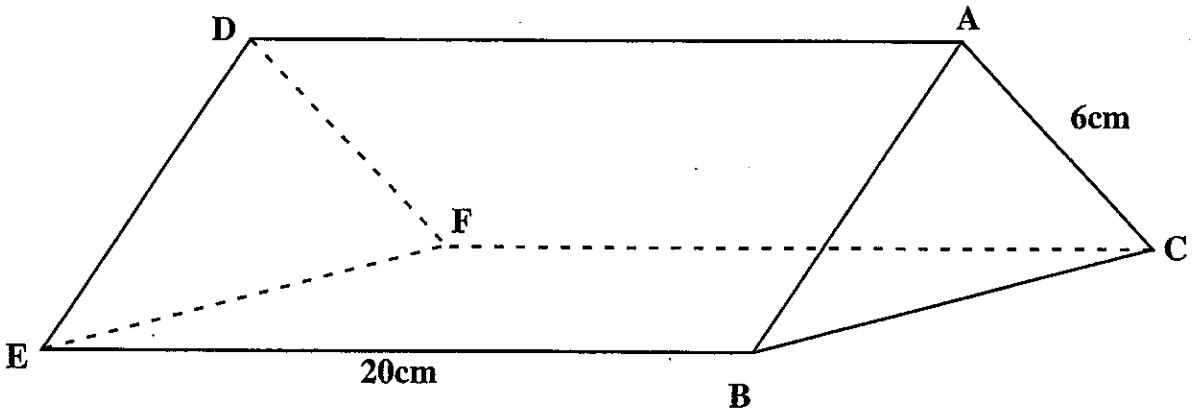

Figure 3

Calculate angle **DCE**. **(8 marks)**

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11. a. A quantity P varies directly as q and inversely as (q^2+1). When $P = 1$, $q = 2$. Express P in terms of q only. **(5 marks)**

b. **Table 2** shows some of values for the equation $y = x^2 + x - 2$.

Table 2

x	-4	-3	-2	-1	0	1
y	10		0	-2	-2	0

(i) Copy and complete the table.

(ii) Using a scale of 2 cm to represent 2 units on the *y-axis* and 2 cm to represent 1 unit on the *x-axis,* draw the graph of $y = x^2 + x - 2$.

(iii) Use the graph to find the minimum value of y. **(10 marks)**

12. a. The sum of the first n terms of a geometric progression, GP, is $2^{n+1} - 1$. Calculate:

(i) the first term of the GP;

(ii) the common ratio of the GP. **(7 marks)**

b. Solve the equation $2 = x^2 + 10x + 21$ giving your answer correct to 2 decimal places. **(8 marks)**

END OF QUESTION PAPER

NB: This paper contains 6 pages.